market_exposed_max_legacy

equity strategy | family: equity-max | status: reference | derived from `market_exposed_max`

What this strategy does

market_exposed with bigger bets for more return: base unit 0.006, 6x leverage cap -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Let net exposure float with the signal; empirically this smuggles in uncontrolled market beta.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.006
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	raw
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	market exposed max legacy	market (CRSP VW)
Annualized return	11.78%	9.94%
Annualized volatility	11.85%	19.35%
Sharpe (NAV-pct)	1.00	0.59
Sortino	1.45	
Max drawdown	-33.0%	-55.5%
Calmar	0.36	
Final NAV ($10M start)	$87,451,313	
Worst day	-7.17%	
% positive days	51.9%	

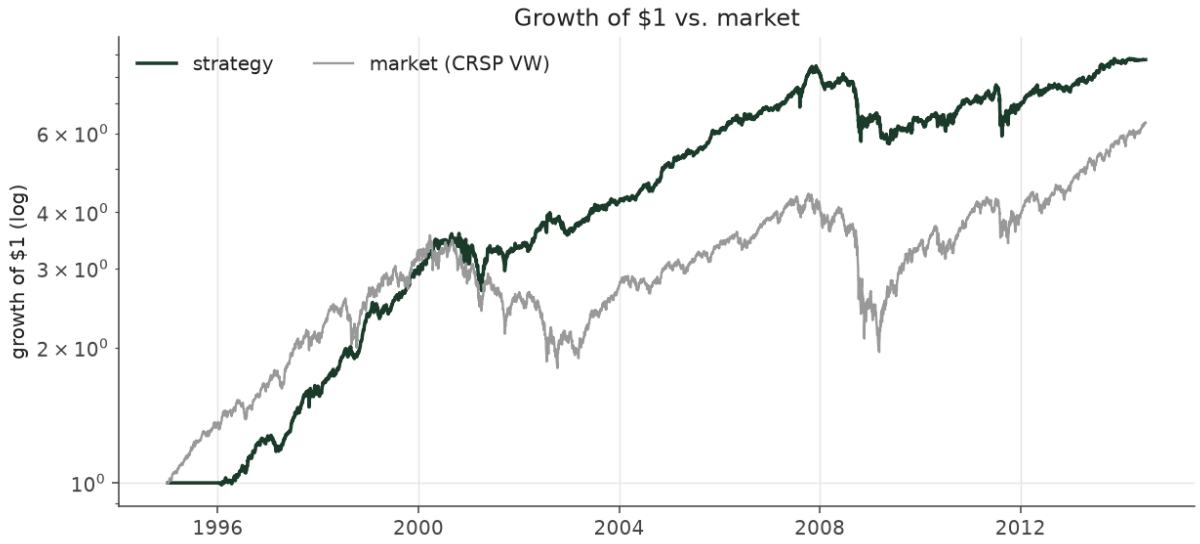
Sharpe 1996-2001	1.64	
Sharpe 2002-2007	1.65	
Sharpe 2008-2013	0.13	

Correlation of daily returns with the market: **0.66**.

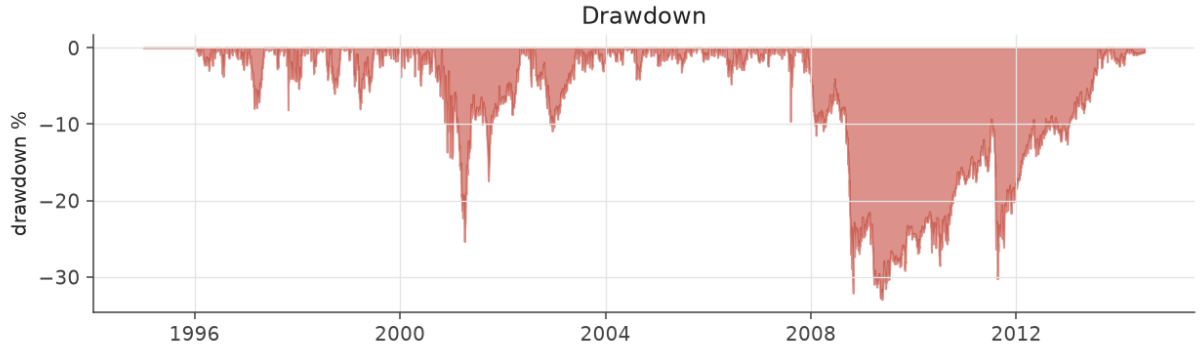
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.999

• best return among equity: `market_boost_max` (0.116); this one: 0.118

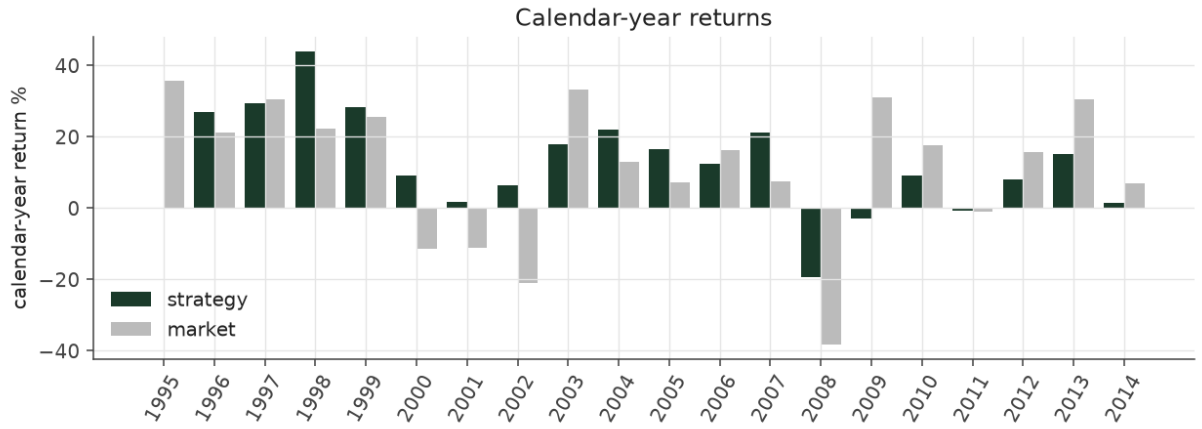
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.357



NAV (log scale)



Drawdown from running peak

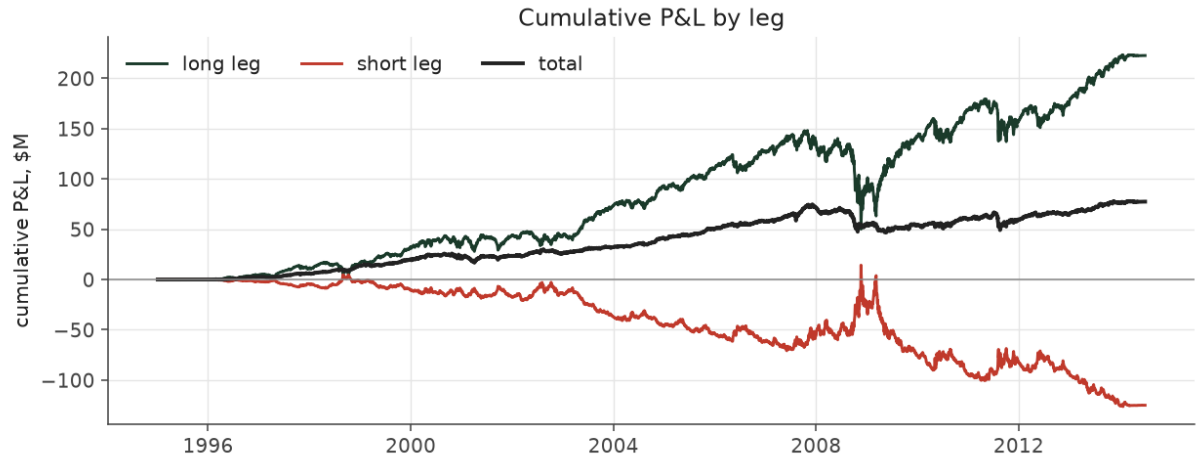


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	222.73	-124.99
P&L after costs ($M)	212.73	-135.28
of which common (average-event) component ($M)	88.61	-80.23

of which excess / PEAD component ($M)	19.82	48.82
excess minus costs ($M)	9.85	38.55
gross return per $ traded (bp)	370.11	-222.37
positions in the money (hit rate)	0.63	0.42
stand-alone Sharpe (leg only)	0.73	-0.46

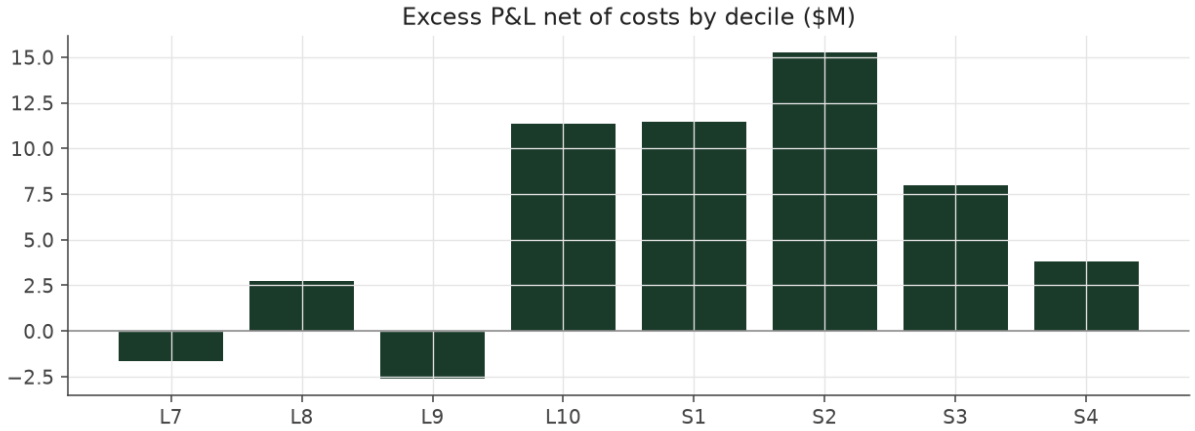
Daily P&L correlation between the two legs: **-0.97**.

• **The short leg loses money in absolute terms** (-125.0M before costs) while the long leg earns +222.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 88.6M / 80.2M. Stripping that common component, the surprise signal itself earned +19.8M on the long side and +48.8M on the short side.

• The two legs' daily P&L are -0.97 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.73 long / -0.46 short vs 1.00 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	514.45	18.09	-0.75	0.88	-1.63	-3.4%	-31.70
long	8	16,588	1,409.31	53.03	5.10	2.38	2.72	5.6%	19.33
long	9	17,747	1,906.30	56.00	0.49	3.11	-2.62	-5.4%	-13.76
long	10	17,174	2,187.84	85.60	14.98	3.63	11.35	23.5%	51.90
short	1	17,169	1,947.43	-49.31	15.10	3.63	11.47	23.7%	58.88
short	2	17,281	1,766.20	-41.46	18.43	3.18	15.26	31.5%	86.37
short	3	15,970	1,363.24	-31.22	10.50	2.50	8.00	16.5%	58.69
short	4	9,171	543.72	-13.28	4.79	0.98	3.81	7.9%	69.97



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	527.66	4.45	7.94	1.84	6.09	12.6%	115.50
2	22,885	1,725.66	9.74	17.59	4.49	13.10	27.1%	75.90

3	25,501	2,640.37	17.24	19.80	5.21	14.59	30.1%	55.25
4	27,583	3,267.00	18.85	13.14	4.94	8.20	16.9%	25.09
5	28,947	3,477.79	27.17	10.17	3.80	6.37	13.2%	18.32

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	300.19	6.42	8.08	1.50	6.58	13.6%	219.04
2	23,084	1,613.35	9.98	17.20	4.84	12.36	25.5%	76.58
3	25,851	2,851.47	15.67	24.14	5.70	18.44	38.1%	64.67
4	27,665	3,422.28	16.70	10.55	4.79	5.76	11.9%	16.84
5	28,874	3,450.23	28.63	8.61	3.45	5.16	10.7%	14.94

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	4,666.77	-12.92	16.72	7.52	9.20	19.0%	19.72
41-60d	25,652	2,632.78	22.99	11.63	4.64	6.99	14.4%	26.55
61-90d	35,324	3,344.89	17.09	18.65	6.09	12.57	26.0%	37.57
>90d	12,298	994.04	50.29	21.63	2.04	19.59	40.5%	197.09

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	1,718.35	23.31	20.69	3.00	17.69	36.5%	102.94
2002-07	39,956	4,116.20	43.81	38.71	7.05	31.66	65.4%	76.93
2008-13	45,278	5,803.94	10.33	9.24	10.24	-1.00	-2.1%	-1.73

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	1,238.86	20.07	5.86	2.33	3.53	7.3%	28.51
Chems	2,590	701.65	6.49	6.16	1.05	5.11	10.6%	72.86
Durbl	2,843	766.16	-0.97	1.45	1.16	0.29	0.6%	3.79
Enrgy	6,194	937.74	5.51	12.22	1.50	10.72	22.2%	114.33
Hlth	11,696	1,162.56	14.23	6.96	2.13	4.83	10.0%	41.57
Manuf	12,034	1,066.72	5.86	5.11	1.91	3.20	6.6%	30.04
Money	17,583	1,062.83	6.91	9.72	2.08	7.64	15.8%	71.87
NoDur	5,081	882.91	8.96	4.82	1.43	3.39	7.0%	38.37
Other	18,377	1,189.85	8.11	8.43	2.35	6.08	12.6%	51.12
Shops	11,984	1,123.54	9.69	7.25	2.00	5.25	10.9%	46.74

Telcm	3,981	796.70	-7.22	-0.34	1.21	-1.55	-3.2%	-19.47
Utils	4,672	708.98	-0.19	1.00	1.15	-0.15	-0.3%	-2.11

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	3,964.97	9.35	14.19	6.36	7.83	16.2%	19.74
low vol	40,091	3,751.90	35.21	29.49	7.09	22.39	46.3%	59.68
mid vol	37,903	3,786.09	32.34	25.15	6.57	18.58	38.4%	49.07

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	11.78%	-0.00%
0.5%	5.5	11.40%	0.38%
1.0%	11.1	11.00%	0.78%
2.0%	22.2	10.11%	1.67%
5.0%	55.4	6.16%	5.62%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.